NAME: _____ Geometry (FL B.E.S.T.)

1.6 Introduction to Proofs

A proof is a chain of true statements, each one justified by a definition, postulate, theorem, or algebraic property — no step is allowed to appear "because it looks true." This section reviews the algebraic properties (reflexive, symmetric, transitive, substitution, addition/subtraction) that fill in most proof reasons, then builds up to full two-column proofs that combine several of them in one argument, such as showing two overlapping segments are equal or formally proving the Vertical Angles Theorem instead of just quoting it.

Practice Set — All Questions

15 questions. Work them on your own paper, then check the key below.

1. Justify $m\angle A = m\angle A$.

2. If $x = 7$, then $7 = x$. Reason?

3. If $a = b$ and $b = c$, then $a = c$. Reason?

4. M midpoint of $\overline{PQ}$. Conclude?

5. $\overline{AB} \cong \overline{CD}$. Conclude about lengths?

6. Vertical angles are always what?

7. Linear-pair angles sum to what?

8. If $m\angle 1 = 40°$ and $m\angle 1 = m\angle 2$, find $m\angle 2$.

9. Expand $2(x + 5)$.

10. If $AB + CD = EF + CD$, conclude $AB = EF$. Reason?

11. Ray bisects an angle. What about the two halves?

12. Is Vertical Angles a postulate or theorem?

13. In a two-column proof, where does "Given" appear?

14. Given: M is the midpoint of $\overline{AB}$, and $AB = 24$. Prove $AM = 12$, giving a reason for each step.

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15. Given: $\angle 1$ and $\angle 2$ are complementary; $\angle 2$ and $\angle 3$ are complementary. Prove $\angle 1 \cong \angle 3$, and name the final property used.

✓ Answer Key

1. Reflexive Property
2. Symmetric Property
3. Transitive Property
4. $PM = MQ$
5. $AB = CD$
6. Congruent
7. $180°$
8. $40°$
9. $2x + 10$
10. Subtraction Property of Equality
11. They are congruent
12. Theorem
13. As the reason for the first statement(s), taken directly from the problem
14. $AM = MB$ (Def. of Midpoint); $AM + MB = AB = 24$ (Segment Addition Postulate); $2AM = 24$ (Substitution); $AM = 12$ (Division Property of Equality)
15. $m\angle 1 + m\angle 2 = 90° = m\angle 2 + m\angle 3$ (Transitive), so $m\angle 1 = m\angle 3$ (Subtraction Property of Equality), so $\angle 1 \cong \angle 3$